BUSINESS ANALYSIS CASE STUDY

StraightTalk Wireless

Prepaid Wireless Services Management Platform

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Prepared By	Yash Isokar | Business Analyst
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1. Executive Summary

StraightTalk Wireless is a prepaid wireless service provider operating under TracFone Wireless, offering no-contract mobile plans across the United States. This case study documents the complete Business Analysis for the StraightTalk Wireless Service Management Platform covering five core customer-facing flows: Activation, Redemption, Deactivation, Reactivation, and Purchase.

The platform enables customers to manage their prepaid wireless services entirely online without requiring interaction with a customer service representative. This document captures all business requirements, functional specifications, user stories with acceptance criteria, and a complete Requirement Traceability Matrix (RTM) across all five service flows.

1.1 Business Problem

Problem Statement

StraightTalk customers face friction in managing prepaid wireless services due to complex multi-step processes, unclear status communication, and inconsistent experiences across service flows. High call center volume (35% of contacts related to activation and redemption issues), customer churn due to failed self-service attempts, and revenue loss from plan purchase abandonment are the core business problems this platform aims to solve.

1.2 Proposed Solution

- Streamlined self-service portal for all 5 service flows
- Real-time status updates and notifications at each step
- Simplified 3-step process for activation and redemption
- Automated reactivation with grace period management
- One-click plan purchase for returning customers

1.3 Business Goals

Business Goal	Target Metric
Reduce call center contacts	Decrease by 40% in 6 months
Improve activation success rate	Increase from 72% to 92%
Reduce plan purchase abandonment	Reduce from 45% to 20%
Improve customer satisfaction	Increase CSAT score by 25 points
Increase reactivation rate	Increase from 55% to 80%

2. Stakeholder Analysis

Stakeholder	Role	Interest / Concern	Influence Level
Product Owner	Sponsor	Platform revenue, customer retention	High
Prepaid Customer	Primary End User	Easy service management, no downtime	High
IT / Dev Team	Builder	Technical feasibility, integrations	High
Network Operations	Service Provider	SIM provisioning accuracy	High
Payment Gateway	External Partner	Secure transaction processing	Medium
Customer Support	Internal End User	Reduced call volume, clear escalation	Medium
Compliance Team	Reviewer	Regulatory compliance, data privacy	Medium
QA Team	Validator	Requirement coverage, defect prevention	Low

3. Project Scope

3.1 In Scope

- Service Activation — New SIM activation and number porting
- Plan Redemption — Airtime card and PIN redemption
- Service Deactivation — Voluntary and non-payment deactivation
- Service Reactivation — Expired account reactivation with grace period
- Plan Purchase — New plan purchase and auto-renewal setup
- Email and SMS notifications for all service state changes
- Real-time service status display on customer dashboard
- Multi-environment validation: SIT, UAT, Production

3.2 Out of Scope

- International roaming plan management
- Family plan or multi-line account management
- Physical store or dealer portal
- Mobile app (iOS/Android) — Phase 2
- BYOD (Bring Your Own Device) compatibility checks

4. Business & Functional Requirements

4.1 Activation Flow

The Activation flow enables new customers to activate their StraightTalk SIM card and select a prepaid plan online without visiting a store or calling customer service.

Req ID	Requirement Description	Priority	Type
ACT-001	Customer shall be able to enter SIM card number (ICCID) to begin activation	High	Functional
ACT-002	System shall validate SIM card number format and availability in real time	High	Functional
ACT-003	Customer shall be able to select a prepaid plan during activation	High	Functional
ACT-004	Customer shall be able to choose a new phone number or port existing number	High	Functional
ACT-005	System shall display estimated activation completion time before confirmation	Medium	Functional
ACT-006	System shall send activation confirmation SMS and email within 5 minutes	High	Functional
ACT-007	System shall display activation status on customer dashboard in real time	High	Functional
ACT-NFR-001	Activation process shall complete within 15 minutes of submission	High	Non-Functional
ACT-NFR-002	Activation page shall load within 3 seconds on standard broadband connection	Medium	Non-Functional

4.2 Redemption Flow

The Redemption flow allows existing customers to redeem airtime cards or PIN codes to add service time or data to their active account.

Req ID	Requirement Description	Priority	Type
RED-001	Customer shall be able to enter a 15-digit PIN code to redeem airtime	High	Functional
RED-002	System shall validate PIN code format, authenticity, and redemption eligibility	High	Functional
RED-003	System shall display current account balance and expiry before redemption	Medium	Functional
RED-004	System shall display new balance and expiry date after successful redemption	High	Functional
RED-005	System shall prevent redemption of already used or expired PIN codes	High	Functional
RED-006	Customer shall receive SMS confirmation of successful redemption immediately	High	Functional

RED-007	System shall log all redemption attempts with timestamp for audit purposes	Medium	Functional
RED-NFR-001	PIN validation and redemption shall complete within 30 seconds	High	Non-Functional

4.3 Deactivation Flow

The Deactivation flow covers both voluntary customer-initiated deactivation and system-triggered deactivation due to non-payment or plan expiry.

Req ID	Requirement Description	Priority	Type
DEA-001	Customer shall be able to request voluntary service deactivation from account settings	High	Functional
DEA-002	System shall display warning message and consequences before deactivation confirmation	High	Functional
DEA-003	System shall trigger automatic deactivation after 60 days of plan expiry	High	Functional
DEA-004	System shall send deactivation warning notification 7 days and 1 day before deactivation	High	Functional
DEA-005	System shall retain customer account data for 90 days after deactivation for reactivation	High	Functional
DEA-006	Customer shall receive deactivation confirmation via email and SMS	High	Functional
DEA-007	System shall release phone number to number pool after 90-day retention period	Medium	Functional

4.4 Reactivation Flow

The Reactivation flow enables customers with deactivated accounts to restore their service within the 90-day retention window.

Req ID	Requirement Description	Priority	Type
REA-001	Customer shall be able to reactivate deactivated account within 90-day window	High	Functional
REA-002	System shall display remaining days in reactivation window on login	High	Functional
REA-003	Customer shall be required to purchase a new plan to complete reactivation	High	Functional
REA-004	System shall restore original phone number if within 90-day retention period	High	Functional
REA-005	System shall send reactivation reminder emails at 30 days, 60 days, and 85 days	Medium	Functional
REA-006	Customer shall receive reactivation confirmation SMS and email within 5 minutes	High	Functional

REA-007	System shall block reactivation attempts after 90-day window with clear error message	High	Functional

4.5 Plan Purchase Flow

The Plan Purchase flow enables both new and existing customers to browse, select, and purchase prepaid wireless plans with multiple payment options.

Req ID	Requirement Description	Priority	Type
PUR-001	System shall display all available prepaid plans with pricing and features	High	Functional
PUR-002	Customer shall be able to filter plans by price, data, and talk/text features	Medium	Functional
PUR-003	System shall support payment via credit card, debit card, and PayPal	High	Functional
PUR-004	Returning customers shall be able to purchase plan using saved payment method in one click	High	Functional
PUR-005	Customer shall be able to enable auto-renewal for selected plan during purchase	Medium	Functional
PUR-006	System shall display order summary with total amount before payment confirmation	High	Functional
PUR-007	Customer shall receive purchase confirmation email with receipt within 5 minutes	High	Functional
PUR-NFR-001	Payment processing shall be PCI DSS compliant	High	Non-Functional
PUR-NFR-002	Payment transaction shall complete within 10 seconds	High	Non-Functional

5. User Stories with Acceptance Criteria

5.1 Activation User Stories

US-ACT-001: SIM Activation

> *As a new StraightTalk customer, I want to activate my SIM card online so that I can start using wireless service immediately without visiting a store.*

Acceptance Criteria:

- Given I am on the activation page, When I enter a valid 19-digit ICCID number, Then the system shall confirm SIM card is valid and available for activation
- Given I enter an invalid ICCID, When I click Continue, Then an error message shall display: 'Invalid SIM card number. Please check and try again'
- Given my SIM is already activated, When I enter my ICCID, Then system shall display 'This SIM is already active' with option to manage existing account
- Given I have selected a plan and entered SIM details, When I confirm activation, Then activation shall complete within 15 minutes and I shall receive SMS confirmation

US-ACT-002: Number Porting

> *As a customer switching from another carrier, I want to port my existing phone number to StraightTalk so that I can keep my current number without disruption.*

Acceptance Criteria:

- Given I select 'Keep my existing number' during activation, When I enter my current number and account details, Then system shall initiate porting request with current carrier
- Given porting is initiated, When I check my activation status, Then system shall display estimated porting completion time (up to 24 hours)
- Given porting is complete, When system processes port, Then I shall receive SMS confirmation: 'Your number has been successfully ported to StraightTalk'
- Given porting fails due to incorrect account details, When system detects failure, Then I shall receive error notification with instructions to retry

5.2 Redemption User Stories

US-RED-001: Airtime Card Redemption

> *As an existing StraightTalk customer, I want to redeem my airtime card PIN online so that I can extend my service without visiting a store.*

Acceptance Criteria:

- Given I am logged in, When I navigate to Redeem Airtime and enter a valid 15-digit PIN, Then system shall validate PIN and display plan details before confirmation
- Given I confirm redemption, When system processes PIN, Then my account service end date shall be extended and new expiry date shall display on dashboard
- Given I enter an already-used PIN, When I click Redeem, Then system shall display: 'This PIN has already been redeemed. Please enter a different PIN'

- Given redemption is successful, When processing completes, Then I shall receive SMS: 'Your StraightTalk service has been extended to [new expiry date]'

5.3 Deactivation User Stories

US-DEA-001: Voluntary Deactivation

> *As a StraightTalk customer, I want to deactivate my service voluntarily so that I can stop charges when I no longer need the service.*

Acceptance Criteria:

- Given I am logged in and go to Account Settings, When I click Deactivate Service, Then system shall display warning: 'Deactivating will end your service. You have X days remaining on your plan'
- Given I confirm deactivation, When system processes request, Then service shall be deactivated and I shall receive confirmation email and SMS
- Given my account is deactivated, When I log in within 90 days, Then system shall display reactivation option with days remaining in retention window
- Given 90-day retention window expires, When system checks account, Then account data shall be permanently deleted and phone number released

US-DEA-002: Automated Deactivation Warning

> *As a customer with an expiring plan, I want to receive advance deactivation warnings so that I have time to redeem airtime or purchase a new plan before losing service.*

Acceptance Criteria:

- Given my plan expires in 7 days, When system runs daily notification check, Then I shall receive email and SMS: 'Your StraightTalk service expires in 7 days. Redeem now to keep service'
- Given my plan expires in 1 day, When system runs notification, Then I shall receive urgent reminder with direct link to redemption and plan purchase pages
- Given my plan has expired and no redemption for 60 days, When system checks, Then account deactivation shall be triggered automatically

5.4 Reactivation User Stories

US-REA-001: Account Reactivation Within Window

> *As a deactivated StraightTalk customer, I want to reactivate my account within 90 days so that I can restore my service and keep my original phone number.*

Acceptance Criteria:

- Given I log in with deactivated account within 90 days, When dashboard loads, Then system shall display: 'Your account is deactivated. You have X days to reactivate and keep your number'
- Given I click Reactivate, When I select a new plan and complete payment, Then service shall be restored within 15 minutes with original phone number
- Given reactivation is successful, When service is restored, Then I shall receive SMS: 'Welcome back! Your StraightTalk service is now active'

- Given I attempt reactivation after 90-day window, When system validates, Then system shall display: 'Reactivation window has expired. A new account and number is required'

5.5 Purchase User Stories

US-PUR-001: New Plan Purchase

> *As a StraightTalk customer, I want to purchase a prepaid plan online so that I can choose the best plan for my needs and pay securely.*

Acceptance Criteria:

- Given I am on the Plans page, When I view available plans, Then all plans shall display price, data allowance, talk and text details, and validity period
- Given I select a plan, When I proceed to payment, Then order summary shall show plan name, price, taxes, and total amount before payment
- Given I complete payment with valid card, When transaction is processed, Then I shall receive payment confirmation email with receipt within 5 minutes
- Given my card is declined, When payment fails, Then system shall display: 'Payment failed. Please check card details or try another payment method' without charging

US-PUR-002: Auto-Renewal Setup

> *As a StraightTalk customer, I want to enable auto-renewal for my plan so that my service is never interrupted due to forgetting to renew.*

Acceptance Criteria:

- Given I am purchasing a plan, When I check the Auto-Renewal option, Then system shall display auto-renewal terms including next charge date and amount
- Given auto-renewal is enabled, When my plan is 3 days from expiry, Then system shall send reminder: 'Auto-renewal of your plan is scheduled for [date]. Amount: $[amount]'
- Given auto-renewal charge fails, When payment is declined, Then system shall notify me immediately and give 3-day grace period before service suspension
- Given I want to cancel auto-renewal, When I go to Account Settings and disable it, Then system shall confirm cancellation and service end date

6. Requirement Traceability Matrix (RTM)

The RTM below maps every business requirement to its corresponding user story, development fix, and test validation across SIT, UAT, and Production environments.

6.1 Activation RTM

Req ID	Requirement	User Story	Dev Fix ID	SIT	UAT	PROD	Remarks
ACT-001	SIM number entry	US-ACT-001	DEV-ACT-001	**Pass**	**Pass**	**Pass**	Verified all formats
ACT-002	SIM validation	US-ACT-001	DEV-ACT-002	**Pass**	**Pass**	**Pass**	Real-time check OK
ACT-003	Plan selection during activation	US-ACT-001	DEV-ACT-003	**Pass**	**Pass**	**Pass**	All plans visible
ACT-004	New number or port existing	US-ACT-002	DEV-ACT-004	**Pass**	**Fail**	**Pending**	Port failure UAT open
ACT-005	Display activation ETA	US-ACT-001	DEV-ACT-005	**Pass**	**Pass**	**Pass**	ETA accurate
ACT-006	Confirmation SMS and email	US-ACT-001	DEV-ACT-006	**Pass**	**Pass**	**Pass**	Within 5 mins verified
ACT-007	Real-time status on dashboard	US-ACT-001	DEV-ACT-007	**Pass**	**Pass**	**Pass**	Status updates live

6.2 Redemption RTM

Req ID	Requirement	User Story	Dev Fix ID	SIT	UAT	PROD	Remarks
RED-001	PIN code entry	US-RED-001	DEV-RED-001	**Pass**	**Pass**	**Pass**	15-digit validated
RED-002	PIN validation and eligibility	US-RED-001	DEV-RED-002	**Pass**	**Pass**	**Pass**	Duplicate PIN blocked
RED-003	Display current balance before redemption	US-RED-001	DEV-RED-003	**Pass**	**Pass**	**Pass**	Balance accurate
RED-004	Display new balance after	US-RED-	DEV-RED-	**Pass**	**Pass**	**Pass**	Expiry

	redemption	001	004				date correct
RED-005	Block used or expired PINs	US-RED-001	DEV-RED-005	**Pass**	**Pass**	**Pass**	Error message clear
RED-006	SMS confirmation on redemption	US-RED-001	DEV-RED-006	**Pass**	**Pass**	**Pass**	SMS within 30 secs
RED-007	Log all redemption attempts	US-RED-001	DEV-RED-007	**Pass**	**Pass**	**Pass**	Audit log confirmed

6.3 Deactivation RTM

Req ID	Requirement	User Story	Dev Fix ID	SIT	UAT	PROD	Remarks
DEA-001	Voluntary deactivation request	US-DEA-001	DEV-DEA-001	**Pass**	**Pass**	**Pass**	Settings flow verified
DEA-002	Warning before deactivation	US-DEA-001	DEV-DEA-002	**Pass**	**Pass**	**Pass**	Warning message clear
DEA-003	Auto deactivation after 60 days	US-DEA-002	DEV-DEA-003	**Pass**	**Pass**	**Pass**	Scheduler verified
DEA-004	Warning notifications 7-day and 1-day	US-DEA-002	DEV-DEA-004	**Pass**	**Pass**	**Pass**	Both reminders sent
DEA-005	Retain data 90 days post deactivation	US-DEA-001	DEV-DEA-005	**Pass**	**Pass**	**Pass**	Data retention confirmed
DEA-006	Deactivation confirmation email and SMS	US-DEA-001	DEV-DEA-006	**Pass**	**Pass**	**Pass**	Both channels verified
DEA-007	Release number after 90 days	US-DEA-001	DEV-DEA-007	**Pass**	**Pending**	**Pending**	UAT in progress

6.4 Reactivation RTM

Req ID	Requirement	User	Dev Fix	SIT	UAT	PROD	Rema

		Story	ID				rks
REA-001	Reactivation within 90-day window	US-REA-001	DEV-REA-001	**Pass**	**Pass**	**Pass**	Window logic correct
REA-002	Display remaining days on login	US-REA-001	DEV-REA-002	**Pass**	**Pass**	**Pass**	Countdown accurate
REA-003	New plan purchase required for reactivation	US-REA-001	DEV-REA-003	**Pass**	**Pass**	**Pass**	Payment flow verified
REA-004	Restore original phone number	US-REA-001	DEV-REA-004	**Pass**	**Pass**	**Pass**	Number restored correctly
REA-005	Reminder emails at 30, 60, 85 days	US-REA-001	DEV-REA-005	**Pass**	**Pass**	**Pass**	All 3 reminders sent
REA-006	Reactivation confirmation SMS and email	US-REA-001	DEV-REA-006	**Pass**	**Pass**	**Pass**	Within 5 mins verified
REA-007	Block reactivation after 90-day window	US-REA-001	DEV-REA-007	**Pass**	**Pass**	**Pass**	Error message clear

6.5 Plan Purchase RTM

Req ID	Requirement	User Story	Dev Fix ID	SIT	UAT	PROD	Remarks
PUR-001	Display all available plans	US-PUR-001	DEV-PUR-001	**Pass**	**Pass**	**Pass**	All plans displaying
PUR-002	Filter plans by price, data, features	US-PUR-001	DEV-PUR-002	**Pass**	**Pass**	**Pass**	Filters working
PUR-003	Payment via card and PayPal	US-PUR-001	DEV-PUR-003	**Pass**	**Pass**	**Pass**	PCI DSS compliant
PUR-004	One-click purchase for returning customers	US-PUR-001	DEV-PUR-004	**Pass**	**Fail**	**Pending**	Saved card issue UAT
PUR-005	Auto-renewal setup during purchase	US-PUR-002	DEV-PUR-005	**Pass**	**Pass**	**Pass**	Renewal logic verified

PUR-006	Order summary before payment	US-PUR-001	DEV-PUR-006	**Pass**	**Pass**	**Pass**	Tax calculation correct
PUR-007	Purchase confirmation email with receipt	US-PUR-001	DEV-PUR-007	**Pass**	**Pass**	**Pass**	Receipt format correct

7. RTM Summary Dashboard

Flow	Total Reqs	SIT Pass	UAT Pass	UAT Fail	PROD Pass	PROD Pending
Activation	9	9/9 (100%)	8/9 (89%)	1 (Port issue)	8/9 (89%)	1
Redemption	8	8/8 (100%)	8/8 (100%)	0	8/8 (100%)	0
Deactivation	7	7/7 (100%)	6/7 (86%)	0	6/7 (86%)	1
Reactivation	7	7/7 (100%)	7/7 (100%)	0	7/7 (100%)	0
Plan Purchase	9	9/9 (100%)	8/9 (89%)	1 (Card issue)	8/9 (89%)	1
TOTAL	**40**	**40/40 (100%)**	**37/40 (93%)**	**2 open**	**37/40 (93%)**	**3**

7.1 Open Defects

Defect ID	Requirement	Description	Severity	Status
DEF-001	ACT-004	Number porting fails for certain carriers in UAT	High	**Fail**
DEF-002	PUR-004	Saved card not loading for returning customers in UAT	Medium	**Fail**

8. Glossary

Term	Definition
ICCID	Integrated Circuit Card Identifier — unique 19-20 digit number identifying a SIM card
PIN Code	Personal Identification Number — 15-digit code on airtime card used for service redemption
Activation	Process of enabling a new SIM card for wireless service on the StraightTalk network
Redemption	Process of adding service time or data to an existing account using an airtime card PIN
Deactivation	Termination of wireless service either voluntarily by customer or automatically by system
Reactivation	Restoration of previously deactivated wireless service within the 90-day retention window
RTM	Requirement Traceability Matrix — document linking requirements to user stories, development, and testing
SIT	System Integration Testing — testing environment where integrated components are validated
UAT	User Acceptance Testing — testing environment where business validates requirements before production
PCI DSS	Payment Card Industry Data Security Standard — security standard for payment card data protection
CSAT	Customer Satisfaction Score — metric measuring customer satisfaction with service experience
Auto-Renewal	Automatic plan renewal using saved payment method before plan expiry to prevent service interruption
Grace Period	90-day window after deactivation during which customer can reactivate and retain original number

9. Document Sign-Off

Role	Name	Sign-Off Date
Business Analyst	Yash Isokar	April 2026
Product Owner	_____	_____
Development Lead	_____	_____
QA Lead	_____	_____

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